Cast Resolver — one cast list per show, before anyone casts a voice

Kuku AI Generalist Assignment · Problem 3 · Devansh · GitHub: https://github.com/devansh265/Personal/tree/main/kuku-cast-resolver

Problem picked, and why

In an AI microdrama every speaker label in the script becomes a voice assignment: once for the original TTS, then once per dub language (kukufm.com lists seven content languages). In the input pack the same character is labelled many ways across 50 episodes: `JULIANA` / `LADY JULIANA`, `GERALD` / `THE FORMER GROOM`, `BLOND NOBLEMAN (Rupert)` / `RUPERT`, `MYSTERIOUS BEARDED MAN` / `THE BEARDED STRANGER` / `ALEXANDER`. 64 labels for roughly 45 speaking roles. Nothing in the pack says who is who; the person casting voices has to read 50 scripts to find out, per show, per language.

Who has it: the dubbing / voice-cast editor, and whoever prompts episode N+1 with no cast bible. **Why it beats P1 and P2:** P1's promo picker is downstream of this (a promo only makes sense alone if names are stable); P2 is outside Kuku's work. This one blocks every dub of every show and is one step of one person's work with one checkable output.

Evidence it is real (from the pack, measured by the script)

Fact	Number	Why it matters
Episodes / dialogue lines (SFX rows excluded)	50 / 1,487	first count was 1,545; 114 were SOUND FX rows the parser treated as a speaker
Distinct speaker labels / speaking roles	64 / ~45	roles counted by hand; the gap is the problem
Aliases that recur *after* the real name is known	4	LADY JULIANA returns in ep 21-24, 34; THE FORMER GROOM in ep 9; BLOND NOBLEMAN up to ep 44 while RUPERT exists from ep 20
Label never named anywhere	LORD, ep 33, 36 lines	unresolvable even by a human: the tool must flag, not guess
Episodes with a different scene-marker format	1 (ep 6)	a strict parser reads zero scenes and says nothing

Pre-reveal descriptors in ep 1-3 (`LADY IN GOLD GOWN`, `THE FORMER GROOM`) are deliberate writing. The drift is that they are never back-filled once the name is known. Ruled out as problems: dub pacing (median 2.5 words/s, p90 3.9), visual description drift (Juliana's eye colour consistent in all 10 mentions).

Assumptions

- Speaker labels drive voice assignment in TTS and dubbing. If casting is already keyed on a bible, the same tool catches writer-side drift instead.
- Likely cause (cannot see the pipeline): each episode is generated with only the previous episode's continuity notes, no cast bible, and no check between the script's label set and the cast list.
- `DOWAGER DUCHESS` (ep 26-28, 43) and `ALEXANDER'S MOTHER` (ep 35-36, 42) are the same woman. No script states it; the tool leaves this to a human.
- External check so far is kukufm.com (seven languages). The Play Store listing would not render, so a viewer or review check is still owed.

How it works — `resolve_cast.py`

Input: a folder of screenplays in the pack format. **Output:** `cast.json` (one row per character: aliases, episodes, lines, drift episodes), `alias_map.csv` (episode × raw label → canonical, with method and confidence), `flags.csv` (what a human must decide), `summary.txt`.

Code, deterministic: parse labels and timings; strip honorifics (`LADY X` → X when X exists); read parenthetical names (`BLOND NOBLEMAN (Rupert)` → RUPERT); detect drift (alias used after the character was named); case-insensitive scene markers so ep 6 does not vanish.

Model, one call per show: the remaining labels, their counts and the camera/action sentence before each line go to a model that returns JSON clusters with confidence and one line of evidence. Never merges two labels that speak to each other. One function, `llm_cluster`, selected by `--provider` and `--model`, so a stronger API is a flag change.

Human: merges under 0.8 confidence and labels with no naming evidence stay in `flags.csv`. `LORD` is flagged, never merged.

Results on the pack (rules only, no API key on the machine at submission)

Metric	Before	After rules	Expected with model
Labels / characters	64 / 64	64 / 60	64 / ~45
Labels carried by the 13 leads (≥20 lines)	17	13 + 4 merged	13
Merges made	0	4 (LADY JULIANA, LADY CATHERINE, LADY CLARISSA, BLOND NOBLEMAN→RUPERT)	~19
Drift episodes surfaced	none visible	4 aliases, 11 episodes	all

Flags for a human	nothing surfaces today	25, with line counts and episodes	~8 (LORD, MAID, JUDGE, …)

Spot check of the 4 rule merges against the scripts: 4/4 correct. An earlier version merged `THE FORMER GROOM` into `CLARISSA` because a parenthetical read "(To Clarissa)"; fixed by taking only a leading lone proper name. Raw outputs are unedited and committed.

Scale and cost

Parsing is free and instant (50 scripts in under a second). The model step is one call per show with ~64 labels and 3 context lines each, roughly 6k input tokens, so a new show costs a few cents on any current model and nothing on a free tier. Ten shows a week is trivial. It runs before TTS, so the fix lands once instead of once per language. For every show Kuku makes: run it on each new episode drop, diff `cast.json` against the previous run, and route only new flags to the editor.

How we would know it worked

- Labels per lead character goes to 1 on every show (today 2.5 on the pack, 17 labels for the leads if counted from 8).
- Editor time to build a show's cast list drops from reading 50 scripts to reviewing one flags list; measure minutes per show for 5 shows.
- Voice swaps for one character after release go to zero; count them from dub QA tickets.
- Flag precision: of the labels flagged, how many did the editor actually need to decide? Target above 70%.

Where it fails

- Deliberate double identities (disguise, twins): the model will want to merge them unless the context says otherwise.
- Unnamed roles with thin context (`LORD`): the right answer is a flag, which still costs human time.
- One-line extras are low confidence by design, so the review list grows with cast size.
- If scripts stop carrying camera/action descriptions, the model loses most of its evidence; the rules still run.
- Rules-only mode misses every descriptor→name case (`THE BEARDED STRANGER` → `ALEXANDER`); those sit in flags until the model runs.

AI tools used, and where they got it wrong

Claude Code for parsing, counting and drafting. Two misses: the first line count included 114 sound-effect rows as speakers; and the first merge rule used any capitalised word in a parenthetical, which turned "(To Clarissa)" into a Gerald→Clarissa merge. Both were caught by reading the output against the scripts, which is the same discipline the tool asks of its human reviewer.

Submission links

- GitHub (prototype, README, raw outputs, this doc): **https://github.com/devansh265/Personal/tree/main/kuku-cast-resolver**
- Raw outputs: the `raw_outputs/` folder in the same repo (cast.json, alias_map.csv, flags.csv, summary.txt). The model request file is withheld because it quotes the confidential scripts.

Appendix — handwritten working notes (GitHub: https://github.com/devansh265/Personal/tree/main/kuku-cast-resolver)

Kuku — AI microdrama

pack: "I Married a Broken Soldier. He was a Duke" · 50 ep · ~2 min each · 1431 dialogue lines (+114 SFX rows)

Making

→ script gen (per episode)

→ TTS voice per speaker label

→ video gen → publish

visual desc drift? Juliana "blue eyes" 10/10 ✓ consistent problem

Promoting

Promo finder = P1 → already their problem

Adapting

→ dub into N languages

→ voice cast per character per language

dub line-fit? median 2.5 w/s, p90 3.9 w/s → room to breathe not it

Speaker labels ≠ characters

65 labels ↑ for ≈45 speaking roles

8 leads carry 20 labels

P2 AI lawyer → not Kuku's work

Pick: cast / speaker-label drift one person · one step · one output you can check

who: dubbing / voice-cast editor (and whoever prompts ep N+1). how I know: pack numbers above + kukufm.com lists 7 languages (Hi, En, Mr, Bn, Te, Ta, Ml) → N = 7. still owed: viewer / review check → empty slot (Play Store blocked fetch)

As-is flow

Prompt + prev-ep continuity notes

↓

Script ep N (labels = what camera knows now)

↓

TTS: 1 voice per label ← who is this?

↓

Video gen

↓

Dub × N languages (cast again, per language)

↓

Publish

one bad label fans out

1 label × 7 languages (kukufm.com) × every episode it recurs in

LADY JULIANA → 104 lines, ep 1-3 **and back in 21-24, 34**

THE FORMER GROOM → ep 1-3, **back in ep 9** after GERALD named ep 4

BLOND NOBLEMAN → 46 lines ep 2→44, = Rupert (parenthetical says so), RUPERT label exists from ep 20

LORD ep 33 → 36 lines, never named anywhere

today: fixed by hand, per show, per language — or not fixed (voice swaps mid-series)

Hypotheses (can't see Kuku's pipeline → all are guesses to test)

H1 — label = camera knowledge, not identity. Ep 1-3 "LADY IN GOLD GOWN", "THE FORMER GROOM" are deliberate pre-reveal descriptors. Bug is they're never back-filled: the descriptor comes back after the name is known (LADY JULIANA ep 21+, FORMER GROOM ep 9).

H2 — honorific style never normalised. JULIANA / LADY JULIANA, CATHERINE / LADY CATHERINE, CLARISSA / LADY CLARISSA. Same person, 2 voices.

H3 — unnamed role never promoted when plot names them. BLOND NOBLEMAN (Rupert) stays a descriptor for 40 episodes. MYSTERIOUS BEARDED MAN / THE BEARDED STRANGER / SCARRED BEARDED MAN → ALEXANDER (ep 8).

H4 — generation sees only the previous episode's continuity notes, no cast bible. So drift accumulates: DOWAGER DUCHESS (ep 26-28, 43) vs ALEXANDER'S MOTHER (ep 35-36, 42) — same woman? plausibly yes, no file says so.

H5 — no check between script labels and TTS/dub cast list. Nothing fails loudly; it ships.

H6 — file format drifts too. Ep 6 writes SCENE END in caps; a simple parser reads 0 scenes and moves on. Silent.

ruled out: dub pacing (p90 3.9 w/s), visual description drift (eye colour stable), promo finding (P1).

RCA results

→ No cast bible across episodes → each episode re-decides who is who

→ Label follows the camera (pre-reveal descriptor) and is never back-filled

→ Honorifics (LADY X vs X) not normalised → 2 voices for 1 person

→ Unnamed role never promoted once the plot names them (Rupert, Alexander)

→ No gate between script label set and TTS / dub cast → ships silently

→ Format drift (ep 6) → parsers drop an episode without error

net: 65 labels → ≈45 people leads: 8 people, 20 labels, ~74% of all lines

person	labels seen	eps
Juliana	JULIANA · LADY JULIANA	1-3, 21-24, 34 as LADY
Alexander	ALEXANDER · MYSTERIOUS BEARDED MAN · THE BEARDED STRANGER · SCARRED BEARDED MAN	3-7 descriptors, named ep 8
Gerald	GERALD · THE FORMER GROOM	1-3, 9 as FORMER GROOM
Clarissa	CLARISSA · LADY CLARISSA · LADY IN GOLD GOWN	1-3, 4-13, 37
Catherine	CATHERINE · LADY CATHERINE	20-50, 27 as LADY
Stepmother	MATRIARCH · NOBLE MATRIARCH · LADY ASHCROFT	1, 3, 4-5
Alexander's mother	DOWAGER DUCHESS · ALEXANDER'S MOTHER	26-28, 43 / 35-36, 42 — **needs human ok**
Rupert	RUPERT · BLOND NOBLEMAN	2-44 / 20+
?	LORD (ep 33)	**unresolvable → flag, don't merge**

Solution — smallest thing: cast resolver

scripts folder (any show)

↓

code: parse labels · count lines · first/last ep · parentheticals "(Rupert)" · who-speaks-to-whom · tolerate caps / format drift

↓

AI: cluster aliases using CAMERA/ACTION context lines → canonical name + confidence + 1-line evidence

↓

human: review merges < 0.8 · unresolvable (LORD ep 33) stays flagged

↓

out: cast.json (id, aliases, eps, lines) · per-ep alias map · flags.csv

why this shape

→ counting is code (free, exact)
→ only the "is X the same as Y" question goes to the model → ~65 labels × 1 call per show, not 1431 lines
→ model swappable: one function, prompt + JSON schema

check it worked

→ labels per lead character → 1
→ spot-check 10 merges vs script
→ dub editor: minutes to build cast per show ↓
→ voice swaps mid-series after release → 0

where it fails

→ twins / disguise plots (same person, 2 identities on purpose)
→ unnamed roles with no context (LORD)
→ minor roles with 1 line → low confidence by design